Code No: 138AC **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, July - 2023
ADVANCED ALGORITHMS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

Used paper July/Aug-2023

PART – A

(25 Marks)

1.a) Explain the best case analysis of heap sort. [2]
b) Define Principle of optimality. [3]
c) Write the general method of Greedy Approach. [2]
d) What is the need of Topological Sorting? [3]
e) Define Zero-one principle. [2]
f) How can you solve systems of linear equations by substitution? [3]
g) Define pattern in string. [2]
h) Write the steps of the Naive String Matching Algorithm. [3]
i) Define the word "Polynomial time". [2]
j) Why the NP-complete problems are significant? [3]

PART – B

(50 Marks)

2.a) Explain the worst case analysis of quick sort with example.
b) Write the role of Algorithms in computing. [5+5]

OR

3.a) What is min heap? Explain heap sort strategy with example.
b) Explain the concept of chain matrix multiplication using dynamic programming. [5+5]

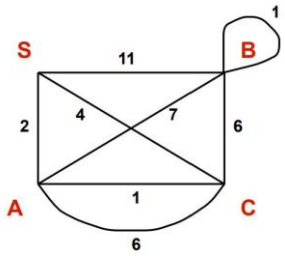
4.a) Apply a variable-length Huffman coding strategy on this frequency character count table.

Character	A	B	C	D	E	F	Total
Frequency	45	13	12	16	9	5	100

b) Write the minimum cost spanning tree prim's algorithm. [5+5]

OR

5.a) Find the Minimum cost spanning tree using Kruskal's algorithm for the given graph.



b) Write the advantages of Maximum Flow algorithm. [5+5]

6.a) Evaluate the time complexity of Strassen's Matrix Multiplication.
b) Explain the concept of Bitonic Sorting Network. [5+5]

OR

7.a) Write the application area of Sorting Network.
b) How to find the Inverse of a 3 × 3 Matrix? [5+5]

8.a) Explain the major steps in Knuth-Morris-Pratt Algorithm.
b) Compare and contrast deterministic and non deterministic finite automata. [5+5]

OR

9.a) Write the brute force algorithm for string pattern matching.
b) How many spurious hits does the Rabin-Karp matcher encounters in Text T = 31415926535....., for string matching, working module q = 11. [5+5]

10.a) What is Travelling Sales person problem? Explain with neat sketch.
b) Explain P, NP and NP-Hard classes with suitable examples. [5+5]

OR

11.a) Explain the strategy to prove that a problem is NP-Hard.
b) What are approximation algorithms? Explain. [5+5]

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